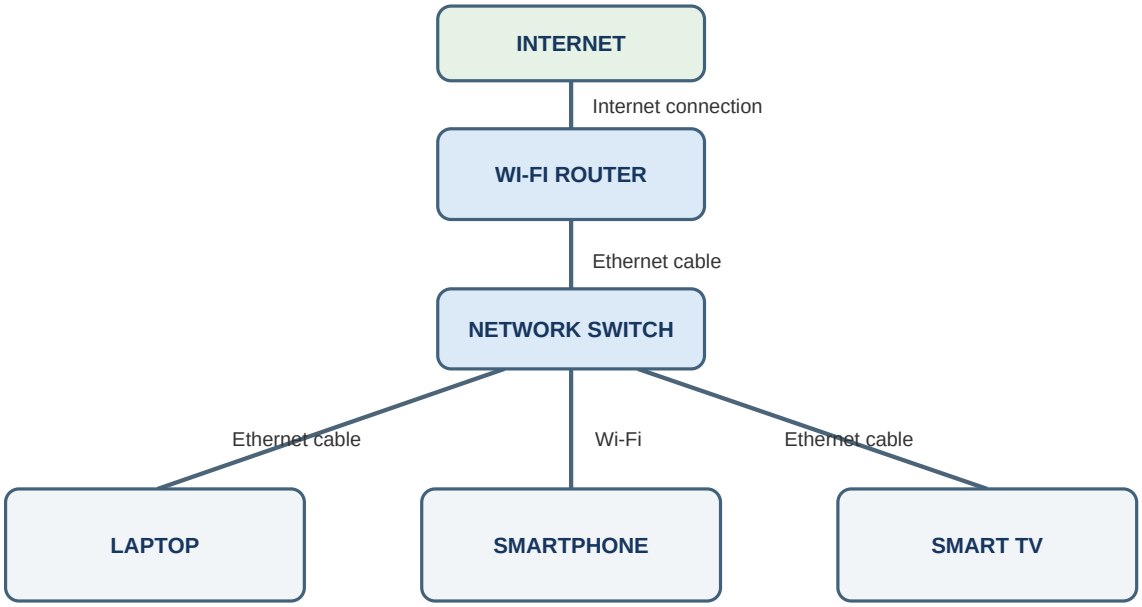
Home Network Devices and Internet Connectivity

Question 1: Draw and label a simple home network diagram showing a router, switch, and three devices.

The diagram below shows a typical home network. The internet connection reaches the Wi-Fi router. An Ethernet cable links the router to a network switch. The laptop and smart TV connect to the switch by Ethernet cables, while the smartphone connects to the router wirelessly through Wi-Fi. A switch adds extra wired ports so several devices can share the same local network.



Example: laptop and TV use wired Ethernet; phone connects wirelessly.

How the connections work: The router connects the home network to the internet and directs traffic between the home network and external websites. The switch connects wired devices in the same local network. The smartphone uses Wi-Fi, so it does not need a cable. In a real home, the router may already have built-in switch ports; a separate switch is useful when more wired ports are needed.

Question 2: Compare at least four network devices and give a real-world use case for each.

Network device	Main function	Real-world use case
Router	Connects different networks and forwards data between the home network and the internet. It commonly provides DHCP, firewall features, and Wi-Fi in one unit.	When you open YouTube or Zomato on your phone, the router forwards your request from the home network to the internet and returns the response.
Switch	Connects wired devices on the same local area network (LAN) and sends Ethernet frames to the correct port using MAC addresses.	In a gaming room, a switch lets several PCs or consoles connect by Ethernet for LAN gaming, game downloads, and file sharing.
Hub	An older, basic device that repeats incoming data to every port instead of sending it only to the intended device. It can create unnecessary traffic.	A hub might appear in an old lab used to demonstrate how shared Ethernet traffic works. Modern homes normally use switches instead.

Network device	Main function	Real-world use case
Wireless access point (AP)	Provides Wi-Fi and bridges wireless devices onto a wired local network. Some routers include an access point inside them.	An AP placed upstairs can provide Wi-Fi coverage for a phone streaming YouTube or a tablet using online classes where the main router's signal is weak.
Wi-Fi extender	Repeats or relays a wireless signal to improve coverage in an area farther from the router. Performance depends on placement and signal quality.	An extender can help a smart TV in a distant room stream videos when the router's Wi-Fi does not reach reliably.

Important note: *A router, switch, hub, and access point do different jobs. Some home products combine several functions in one box, but the functions remain different.*

Question 3: Explain how data travels from your phone to the internet when you order food on Zomato.

Imagine that you are at home and use the Zomato app to order dinner. The phone first connects to the home Wi-Fi network. The wireless access point function in the router sends your phone's data onto the local network. The router then forwards the request through your internet service provider (ISP) towards Zomato's servers. Zomato processes the request and sends information such as restaurant menus, prices, order confirmation, and delivery updates back through the internet to your router and then over Wi-Fi to your phone.

Step-by-step journey of the order

Phone and Wi-Fi: You open Zomato, select food, and tap Place Order. The phone creates network data and sends it wirelessly to the router's access point.

Access point: The access point receives the Wi-Fi frames from the phone and bridges the data onto the home network. It provides the wireless connection; it does not decide the route across the internet.

Switch (if present): If the router is connected to a separate switch, the switch forwards the wired Ethernet traffic to the router's port. If the home setup has no separate switch, this step is handled by the router's built-in Ethernet switching function.

Router: The router checks where the data must go and forwards it towards the ISP. It commonly uses network address translation (NAT) so multiple home devices can share one public internet connection.

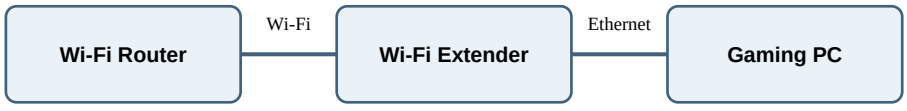
Internet and Zomato servers: The ISP and other internet networks carry the request to Zomato's online services. The app uses secure HTTPS communication to protect information in transit.

Response returns: Zomato sends the response back through internet networks to the home router. The access point transmits it over Wi-Fi to the phone, and the app displays the order confirmation and updates.

The same process happens when the app refreshes restaurant menus or shows delivery status. The exact network path may pass through many ISP routers on the internet, but at home the main roles are Wi-Fi access, local forwarding where a switch is used, and routing to the ISP.

Question 4: The Wi-Fi router is far away, and the gaming PC only has an Ethernet port. Suggest two devices to connect the PC to the internet.

Option A — Wi-Fi extender with an Ethernet port

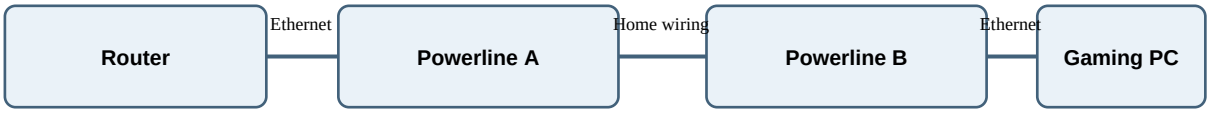


PC plugs into extender with an Ethernet cable

A Wi-Fi extender is placed somewhere between the router and the gaming PC, where it can still receive a strong Wi-Fi signal. The extender connects wirelessly to the router, and the gaming PC connects to the extender using an Ethernet cable. This is useful when running a long cable is difficult. For gaming, place the extender carefully: a weak wireless link can still cause lag or packet loss, and some extenders add latency.

Setup steps: Plug in the extender near the router for initial setup; pair it with the router's Wi-Fi using the manufacturer's app or setup page; move it to a location with a good signal; connect the PC's Ethernet port to the extender's LAN port; then test internet access and game latency.

Option B — Powerline Ethernet adapter kit



Two adapters plug into wall outlets; electrical wiring carries the network signal.

A powerline kit normally uses two adapters. Connect the first adapter to the router with an Ethernet cable and plug it directly into a wall outlet. Plug the second adapter into a wall outlet near the gaming PC and connect it to the PC with another Ethernet cable. The adapters send network data through the home's electrical wiring. This can provide a wired connection without running a long Ethernet cable through the house.

Setup steps: Plug adapter A into a wall outlet near the router and connect it to a router LAN port; plug adapter B into a wall outlet near the PC; pair the adapters using their Pair buttons if required; connect adapter B to the PC by Ethernet; then check the connection. Avoid power strips or surge protectors if the adapter manufacturer advises against them. Speed depends on the home's wiring, circuit layout, and electrical noise.

> ***Which option is better?*** *A powerline kit can be a good first choice if the home's electrical wiring supports it, because the PC gets a wired Ethernet link. A Wi-Fi extender with an Ethernet port is easier where the Wi-Fi signal is strong but the room has no Ethernet cable. Neither guarantees the lowest latency in every home; test the connection and compare ping, packet loss, and speed.*

Conclusion

A home network uses different devices for different tasks. The router connects the home network to the internet, a switch expands wired connections, a hub repeats traffic to all ports, and an access point provides Wi-Fi. For a distant gaming PC that only has Ethernet, a Wi-Fi extender with an Ethernet port or a powerline adapter kit can provide internet access without a direct long cable to the router.